

HAYOUNG DOO

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🏠 <https://doouv.github.io/profile/>

📄 <https://github.com/doouv>

RESEARCH INTERESTS

AI for Materials, Self-driving Laboratory, Physics and experiment-aware AI, decision-making intelligence

EDUCATION

Ewha Womans University, Seoul, South Korea
M.S. in Chemical Engineering and Materials Science

Sep. 2024 – Aug. 2026
GPA 4.30/4.30

- Thesis: “Equivariant Latent Diffusion Framework for Inverse Design of Heterogeneous Catalysts”

Ewha Womans University, Seoul, South Korea
B.S. in Chemical Engineering and Materials Science

Mar. 2020 – Aug. 2024
GPA 3.75/4.30

- Magna Cum Laude, Dean’s List for Four Semesters
- Thesis: “Machine-learning-driven electrocatalyst design: Open Catalyst Project and recent trends”

RESEARCH EXPERIENCE

Research Associate, Institute for Multiscale Matter and Systems (IMMS),
Ewha Womans University, Seoul, ROK

Sep. 2026 – Present

- Research in AI for materials as part of the NRF-funded NRL 2.0 program, with Prof. Jonggeol Na.

Undergraduate and Graduate Researcher, Ewha Womans University,
Seoul, South Korea.

Sep. 2024 – Aug. 2026

- Generative inverse design of heterogeneous catalysts through the NRF-funded Supercatalyst Data HUB project; contributed to AI-agent research under the NRL 2.0 program, with Prof. Jonggeol Na.

Undergraduate Researcher, Korea Institute of Science and Technology (KIST),
Seoul, South Korea.

Mar. 2020 – Aug. 2024

- Machine-learning interatomic potentials and molecular dynamics for solid electrolytes, with Dr. Byungju Lee.

PUBLICATIONS

JOURNAL & CONFERENCE PAPERS

† Equally Contributed, * Corresponding Authors

- [c.1] Ahn, G.†, Lee, D.†, **Doo, H.**, Na, J.* (2026). MIND: AI Co-Scientist for Material Research. *PKDD 2026 (Demo Track)* [arXiv:2604.13699v1](https://arxiv.org/abs/2604.13699v1)
- [j.1] Kim, Y.†, **Doo, H.†**, Shin, D.†, Lee, S. Y.†, Roh, Y.†, Park, S., ... & Na, J.* (2025). Self-driving laboratories with artificial intelligence: An overview of process systems engineering perspective. *Computers & Chemical Engineering*, 109266. <https://doi.org/10.1016/j.compchemeng.2025.109266>

PREPRINTS

- [j.2] **Doo, H.†**, Mok, D.†, Back, S.*, Na, J.* (2026). Catalyst Diffusion Transformer: Generative Inverse Design of Heterogeneous Catalysts. [arXiv:2607.24272](https://arxiv.org/abs/2607.24272) [submitted]

CONFERENCE ACTIVITIES

ORAL & POSTER PRESENTATIONS

- [o.1] “Catalyst Diffusion Transformer: Generative Inverse Design of Heterogeneous Catalysts”
AIChE Annual Meeting | Nov. 2026, Minneapolis, USA
- [p.1] “Equivariant latent diffusion for the generation of oxygen evolution reaction catalysts”
AIChE Annual Meeting | Nov. 2025, Boston, USA
- [p.2] “Foundation model to accelerate oxygen evolution reaction catalyst discovery”
KICChE Annual Meeting | Apr. 2025, Daegu, South Korea
- [p.3] “Autonomous chemical reaction optimization through derivative-free optimization”
KICChE Annual Meeting | Oct. 2024, Busan, South Korea

PARTICIPATION

- [1] AI4X-Accelerate Conference | Jun. 2026, Singapore
- [2] Nature Conference: AI for Materials, Materials for AI | Feb. 2025, Daejeon, South Korea
- [3] Symposium of Multi-scale Modeling and Materials Research based on Machine Learning Potential | Jun. 2023, Daejeon, South Korea

SCHOLARSHIPS

- Master's Excellence Scholarship in Science and Engineering** | KOSAF 2025 – 2026
- AI Seoul Tech Scholarship** | Seoul Scholarship Foundation 2025
- Graduate Admission Scholarship** | Ewha Womans University 2024 – 2025
- Full tuition for one year; Granted for high achievement in undergraduate coursework.
- Dooeul Scholarship** | Dooeul Scholarship Foundation 2021 – 2024
- Full tuition for three years; Granted for high-achieving female students with leadership potential, an international perspective, and professional excellence.

HONORS & AWARDS

- Best Poster Presentation Award**, BK 21 Workshop | Ewha Womans University Dec. 2025
- Finalist**, LG Chem-KICHe Petrochemical Olympiad | LG Chem & KICHe Oct. 2022
- 3rd Place**, Undergraduate Capstone Design Competition | KICHe Oct. 2021

TECHNICAL SKILLS

- Programming & Systems** Python, Docker, Git, Linux, Bash
- ML Training** PyTorch, PyTorch Lightning, Hydra, Weights & Biases
- Scientific Computing** ASE, pymatgen, matminer, OVITO, VESTA

TEACHING & ACADEMIC SERVICE

- Lab Server Administrator** | AMSL, Ewha Womans University Jan. 2026 – Present
- Managed shared GPU and server environments for multi-user laboratory research, including Docker-based and Slurm-managed cluster resources.
- Teaching Assistant, Chemical Engineering Mathematics** | Ewha Womans University Mar. 2025 – Jun. 2025
- Conducted in English; Handled student inquiries, graded assignments and exams
- Teaching Assistant, 2025 Process Systems Engineering Summer School** | KICHe Aug. 2025
- Prepared lecture notes and tutorial code on reinforcement learning and diffusion models for beginner-level educational sessions; materials were shared via GitHub.

ADDITIONAL TRAINING

- UR Collaborative Robot Expert Training Course** | DAS Korea, Suwon, South Korea Jan. 2026
- Received hands-on training in collaborative robot operation for experimental automation, including pick-and-place tasks, conveyor belt integration, and 3D simulation.

OUTREACH & LEADERSHIP

- Hyundai Motor Group Education Volunteer Corps (8th)** | Seoul, South Korea Jan. 2022 – Jan. 2023
- Tutored and provided emotional support to children at local community centers.
- E-Chem Academic Club (Chemical Engineering Student Organization)** Jul. 2022 – Feb. 2023
- Served as president and organized major-related academic activities.
 - Led journal club discussions on topics including deep learning in the semiconductor industry.